

Insertion Sort

Insertion sort is a simple sorting algorithm that builds the sorted array one element at a time.

It takes one element from the unsorted part and inserts it into its correct position in the sorted part of array.

Working Principle

- Assume the first element is already sorted.
- Pick the next element called key.
- Compare the key with the element on its left.
- Shift all larger elements one position to the right.
- Insert the key at its correct position.
- Repeat until the entire array is sorted.

Algorithm

Step-1 Assume the first element is already sorted
 $i = 1;$

Step-2 Store the current element as the key

$$\text{key} = \text{arr}[i];$$

Step-3 Compare the key with previous element

$$j = i - 1;$$

Step-4 while the previous element is greater than the key, shift it one position to the right.

```

while (j >= 0 && arr[j] > key)
{
    arr[j+1] = arr[j];
    j--;
}

```

Step-5 insert the key into its correct position

$arr[j+1] = key;$

Org arr 13 46 24 52 20 9

Pass 1

key = 46

13 | 46 24 52 20 9
 $13 < 46$ No shifting

Pass 2

key 24

13 46 | 24 52 20 9
 $46 > 24 \rightarrow$ shift 46

13 24 46 52 20 9

Pass 3

key = 52

13 24 46 | 52 20 9

52 is already greater than 46 $\therefore$ no shifting

Pass 4

key = 20

Shift :

52 $\rightarrow$ right46 $\rightarrow$ right24 $\rightarrow$ right

insert 20

13 20 24 46 52 | 9

Pass 5

key = 9

Shift :

52 $\rightarrow$ Right46 $\rightarrow$ Right24 $\rightarrow$ right20 $\rightarrow$ right13 $\rightarrow$ right

insert 9

9 13 20 24 46 52.

Program:-

```
int main ()
{
    int arr [6] = {13, 46, 24, 52, 20, 4};
    for (int i = 1; i < 6; i++)
    {
        int temp = arr [i];
        int j = i - 1;
        while (j >= 0 && arr [j] > temp)
        {
            arr [j+1] = arr [j];
            j--;
        }
        arr [j+1] = temp;
    }

    cout << "Sorted Array :- ";
    for (int i = 0; i < 6; i++)
    {
        cout << arr [i] << " ";
    }

    return 0;
}
```